OpenWebUI on Azure

A repeatable, closed-environment foundation for client chatbots

Architecture · Infrastructure · Delivery · Ways of working

~15 min + Q&A

Note — this is the v1 design-review deck

It captures the original architecture we walked through with Andreas.
Since then the implementation has evolved:

- **Model layer:** Azure OpenAI (GPT-4o) → **Azure AI Foundry** with per-client mix of **Claude Sonnet/Haiku 4.5 (MaaS)** and/or **Azure OpenAI GPT-4o/mini**; `app/models.yaml` ∩ `foundry_deployments` renders the LiteLLM config per-client at Terraform apply time
- **Digest summariser:** single prompt → **tool-calling Claude Haiku 4.5 agent** (`get_chat_titles`, `get_usage_stats`), full trace captured in Langfuse
- **RAG:** OpenWebUI's built-in → **custom ingestion pipeline** (Blob → extract → `RecursiveCharacterTextSplitter` → `text-embedding-3-large` via LiteLLM → pgvector `HALFVEC(3072)` with HNSW / `halfvec_cosine_ops`)
- **New:** TypeScript/React admin UI on a dedicated Front Door endpoint (Entra SSO, Langfuse-backed dashboard, runtime config served from `/api/config`)
- **New:** Managed DevOps Pool stack (`terraform/stacks/ci_pool/`) for VNet-injected ephemeral CI agents

Current architecture diagram + operational notes in the repo `README.md`.
The rest of this deck is retained for the original reasoning and trade-offs.

1 · Framing

The ask: a generic chatbot foundation, deployable per client, that becomes the base for their AI initiatives.

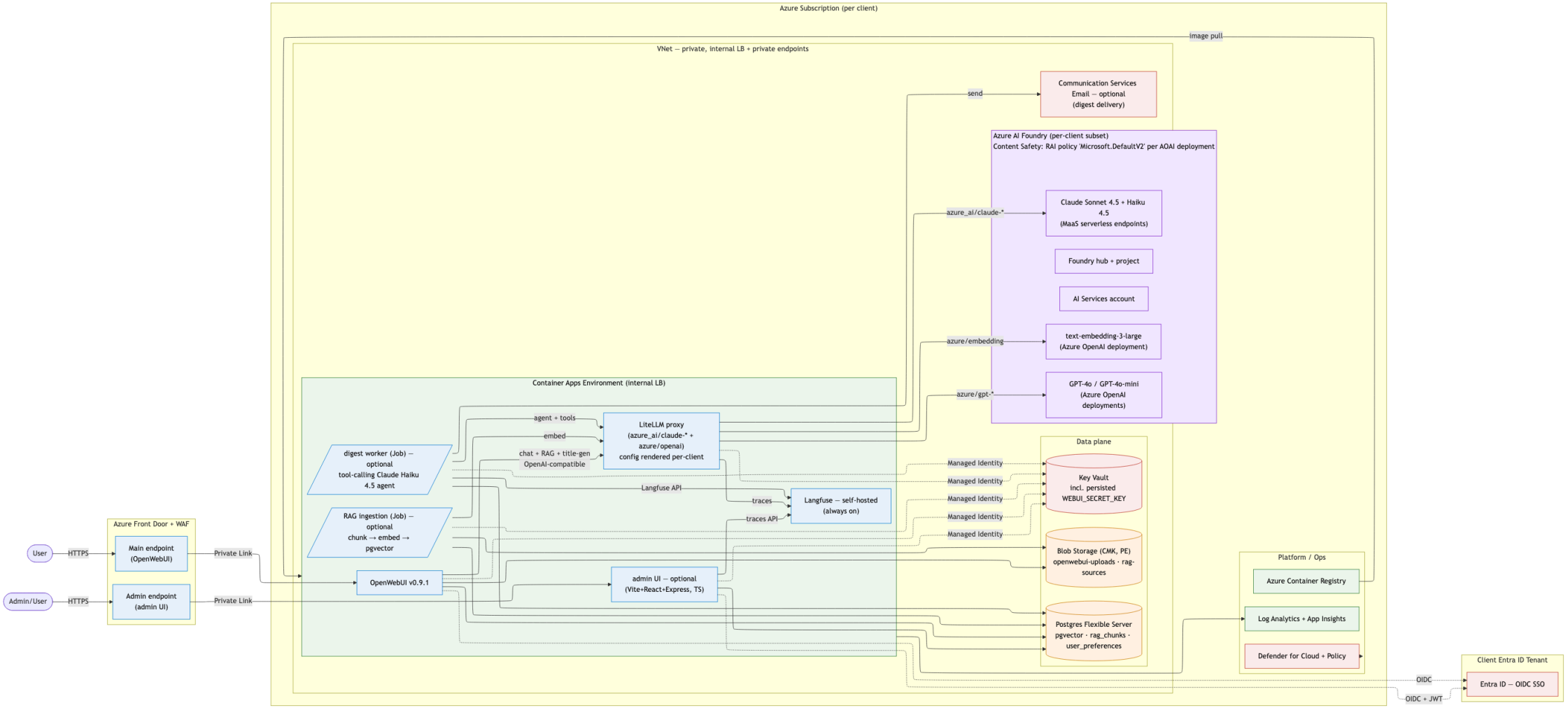
Constraints:

- Runs entirely on Azure — closed, secure data plane
- Infra + app fully scripted
- CI/CD for infra changes, new models, and bug fixes

My design stance:

- Boring defaults, platform-native services
- One template → many clients, parameterised not forked
- Optimise for *iteration speed on prompts and models*, not infra heroics

2 · Target architecture



2 · Architecture — component choices

Layer	Choice	Why this, not that
Ingress	**Front Door + WAF**	TLS, OWASP, DDoS; private origin
Identity	**Entra ID OIDC** + Managed Identities	SSO per client tenant, no app secrets
Compute	**Azure Container Apps** (internal, VNet)	Right-sized; revisions + canary; no K8s tax
State DB	**Postgres Flexible Server** (private)	OpenWebUI first-class support
Vectors	**pgvector** → escalate to **Azure AI Search**	Start simple; upgrade when retrieval demands
LLM	**Azure OpenAI** + **LiteLLM** proxy *(shipped as: Azure AI Foundry hosting Claude MaaS + AOAI, behind LiteLLM)*	One endpoint for OpenWebUI, many models behind it
Files	**Blob Storage** (private endpoint, CMK)	RAG sources + uploads
Secrets	**Key Vault** + Managed Identity	Zero secrets in pipelines/app
Obs (infra)	**Log Analytics + App Insights**	Native, alert rules as code
Obs (LLM)	**Langfuse** (self-hosted Container App)	Traces, token cost, eval gates
Network	VNet-injected Container Apps + Private Endpoints (DB, Blob, KV, Foundry)	Closed data plane; public endpoints disabled on backing services

2 · Architecture — key trade-offs

Container Apps vs AKS

- AKS only if the client already runs K8s. Otherwise Container Apps wins on ops cost and time-to-first-deploy.

pgvector vs Azure AI Search

- pgvector: one less service, good enough for most corpora.
- AI Search: hybrid BM25+vector, semantic ranker, skillsets. Worth the jump when retrieval quality becomes the bottleneck.

Single-tenant per client vs shared multi-tenant

- Recommend **per-client stack**. Data isolation, blast-radius control, per-client compliance posture. IaC makes the duplication cheap.

Azure OpenAI only vs LiteLLM in front

- LiteLLM adds one hop, buys us multi-model routing, per-user quotas, and a clean seam for adding non-Azure models later if policy allows.

2 · Rejected alternatives

From the brief: *"alternatives are welcome"*. Here's what I considered and chose against — every row is a **per-client decision**, not a platform absolute.

Rejected	Chose	Why not	Revisit when
AKS	Container Apps	K8s ops cost doesn't pay back for stateless containers; no team muscle to amortise	Client already runs K8s at scale
Bicep	Terraform	No technical win; you already run Terraform — switching costs are real, not theoretical	Client mandates Microsoft-native IaC
Pulumi	Terraform	Overkill for this scope; HCL + modules is enough	TS-first team with heavy contract-testing needs
Shared multi-tenant	Per-client stack	Data isolation, blast-radius control, per-client compliance posture; IaC makes duplication cheap	Many small clients where per-client cost dominates
Azure AI Search (day 1)	pgvector	One less service, good enough for most corpora, one-module swap when it isn't	Retrieval quality is the bottleneck; hybrid BM25 + vector needed
AOAI direct (no LiteLLM)	LiteLLM in front	One OAI-compatible seam → multi-model routing, per-user quotas, non-Azure optionality — worth the extra hop	Single model forever, no quota policy, no non-Azure ambition

Stance: boring defaults, escape hatch named on every line.

3 · Infrastructure as Code

Terraform, matching the existing ADO setup.

```
terraform/
├── modules/            # network, container-app, postgres, openai, langfuse, frontdoor
├── stacks/openwebui/   # the composed product
└── envs/
    └── <client>/
        ├── dev.tfvars
        ├── test.tfvars
        └── prod.tfvars
```

- **Remote state** in Azure Storage, one file per client/env, blob-lease locking
- **Naming + tagging** via `azurecaf` + Azure Policy; every resource traces to client/env/cost-centre
- **Guardrails** at subscription level (Defender for Cloud, Policy initiatives) — not re-implemented per stack
- Alternatives considered: **Bicep** (fine, but no reason to switch); **Pulumi** (overkill here)

4 · CI/CD — two pipelines, trunk-based

Infra pipeline (`terraform/**`)

1. PR → `fmt` · `validate` · `tflint` · `checkov` · `plan` as PR comment
2. Merge `main` → apply `dev` → approval → `test` → approval → `prod`
3. Same pipeline, per-client variable group, one ADO environment per client-env

App pipeline (`app/**`)

1. Build OpenWebUI container (pinned upstream + our config layer)
2. Push to **ACR** tagged with commit SHA, SBOM attached, **Trivy** scan gate
3. Deploy via **Container Apps revision** with traffic split (10 % → 100 %), auto-rollback on health-probe failure

Model changes as config

- `models.yaml` → PR → pipeline validates against live Foundry / AOAI quota
- Terraform renders `active models = models.yaml ∩ client.foundry_deployments` into `LITELLM_CONFIG_YAML` at apply time — **no image rebuild** for model additions/retirements

5 · Security & compliance

- **Private-only data plane.** No public endpoints on DB, Foundry / AOAI, Blob, Key Vault (PE-gated; cross-region Foundry deployments fall back to service firewall).
- **Entra ID SSO** into OpenWebUI and the admin UI; managed identity on every container.
- **WAF in Prevention mode** (DRS 2.1 + Bot Manager); narrow per-client exceptions for OpenWebUI chat-payload and signup-avatar false positives.
- **Content Safety** on prompts + responses; prompt-injection guardrails enforced by Langfuse evals in CI.
- **Audit**: diagnostic settings → Log Analytics → optional SIEM export per client.
- **Data residency**: region pinned per client; CMK on Storage + Postgres when required.
- **Supply chain**: pinned upstream image digest (OpenWebUI `v0.9.1`), Trivy HIGH/CRITICAL gate on every build *plus* a pre-merge `trivy fs` scan over `app/`, `scripts/`, `.azuredevops/` (vuln + misconfig + secret), non-root containers across the board, SBOM attached, signed commits.

6 · Observability & cost

Infra / app health

- Azure Monitor workbooks, alert rules as code, SLOs per service.

LLM layer — the one that actually matters

- **Langfuse** (self-hosted Container App) for traces, token cost, latency, eval datasets — LiteLLM, digest agent, RAG embeddings, and admin UI all feed it
- Prompt iteration becomes data-driven: compare model/prompt versions before promotion
- Eval suite runs in CI (`tests/eval/`); failing evals block prod deploys when the eval stage is enabled

Cost

- Budget alerts per client subscription
- Per-user/-group token quotas in LiteLLM
- Container Apps scale-to-zero on non-prod

7 · Ways of working

- **Template repo** per product: `openwebui-azure-template`
  - New client = fork + tfvars + ADO variable group → first deploy in a day
- **Definition of done** per PR: plan clean · image scanned · dev deployed · Langfuse evals green
- **Model onboarding is a PR**, not a ticket — reviewer checks quota, cost, content-safety config
- **Runbooks live next to the code.** Postmortems drive module changes, not tribal knowledge.
- **Upstream tracking**: pin OpenWebUI version, test in dev weekly, promote monthly

8 · What I'd validate in week 1

- Client's existing Azure landing zone — plug into their hub or build standalone?
- Regulatory asks — CMK? data residency? log retention? DPA?
- Expected user count + model mix — drives Container Apps scale rules and AOAI quota requests
- SSO source of truth and group model

These answers change 2–3 cells in the trade-off table. Nothing below.

9 · Future possibilities

Optionality — the stack is architected to absorb each of these without redesign. Not on the roadmap, called out as extension points.

Cost governance in CI

- **Infracost** in `infra.yml` → per-PR monthly-cost delta as a PR comment, alongside the existing `plan` output. Cost visible *before* merge, not after the bill.
- Budget policy as code — Azure Policy blocks SKUs above a threshold without explicit override.

Workflow / automation layer

- **n8n** as a Container App inside the VNet — low-code flow builder, calls LiteLLM for LLM steps, Langfuse for traces.
- Non-devs compose agents ("inbox → summarise → post to Teams") without writing Python. Same Entra SSO, same cost seam, same observability as the chatbot.
- Directly answers *"foundation for the client's AI initiatives"* — the chatbot is v1, the workflow plane is v2.

Capability growth

- **RAG-as-tool** — wire the custom retrieval path into OpenWebUI as a callable tool so chat grounds on the corpus automatically (today it's CLI + agent-importable).

Thank you

Questions / deeper dive on any layer?

Architecture · IaC · CI/CD · Security · Observability · WoW